



Are there any OER resources, textbooks , materials for a college course that covers content as is noted in the following course description:

" introduction to the ethical and legal issues pertaining to network and computer security testing. Students will explore an introduction to the hacking methodology and network penetration testing using industry-recognized tools. Hands-on assignments will be used to help students develop introductory technical skills relevant to entry-level cybersecurity professionals, including demonstration of tools that can be used to gain information about a computer network, web applications, and databases."

Yes, there are OER (Open Educational Resource) textbooks and materials that fit a college course covering ethical and legal issues in network and computer security testing, including hands-on hacking methodology and penetration testing with industry tools.

Main OER Textbook

- **Ethical Hacking: Network Security and Penetration Testing** (Galileo OER Collection)
 - This free, peer-reviewed textbook provides a practical introduction to ethical hacking, legal and ethical considerations, and step-by-step penetration testing.^{[1] [2]}
 - Covers topics such as the hacking methodology, reconnaissance, scanning, vulnerability exploitation, post-exploitation, reporting, and the ethical/legal frameworks governing penetration testing.
 - Includes labs and assignments that focus on technical skills with common tools, such as Nmap, Wireshark, Metasploit, and others, aligning with skills needed for entry-level cybersecurity professionals.

Other Open Materials and Courses

- **Practical Ethical Hacking Free Series (TCM Security)**
 - A set of free, hands-on video courses, including introductions to hacking tools, penetration testing methodology, web application hacking, and OSINT. These are suitable for beginners and can be integrated into college curricula.^[3]
- **Open Educational Resource Lists (ASCCC OERI, WCCnet, Kennesaw State, GitHub)**
 - University-driven OER listings and GitHub resource lists aggregate free textbooks, lab exercises, and online labs covering cybersecurity, hacking, and penetration testing.^[2]
^{[4] [5] [6]}

Free Tools and Labs

- **Free Penetration Testing Tools**

- Several platforms offer open-access tools for reconnaissance, network mapping, web vulnerability scanning, and information gathering (e.g., Nmap, Burp Suite Community, OWASP ZAP). These tools are commonly used in OER lab assignments and exercises. ^[7]
^[8] ^[9] ^[10]

- **Online Cybersecurity Labs & DIY Platforms**

- Links to practice platforms such as Hack The Box, TryHackMe, Portswigger's Web Security Academy, and IppSec's YouTube tutorials, which offer hands-on network and web application security testing experience at no cost. ^[11]

Suitability

These resources fully address course requirements for:

- Exploring ethical and legal frameworks for penetration testing and security auditing, ^[1]
- Teaching the hacking methodology in a structured, legal, and ethical context,
- Developing technical skills in information gathering, scanning, exploitation, and reporting,
- Using industry-recognized tools in guided, hands-on assignments.

All are freely available online and supported by supplementary materials for both instructors and students. Yes, there are multiple OER (Open Educational Resource) textbooks, courses, and lab materials suitable for a college course on ethical and legal issues in network and computer security testing with hands-on coverage of hacking methodology and industry tools.

OER Textbooks

- **"Ethical Hacking: Network Security and Penetration Testing" (Galileo OER/Affordable Learning Georgia)**

This peer-reviewed textbook covers the ethical, legal, and technical aspects of hacking and security testing. It provides foundational knowledge, introduces professional penetration testing tools (like Nmap, Wireshark, and Metasploit), and includes hands-on assignments and labs that align directly with entry-level cybersecurity skills. ^[2] ^[1]

Free Hands-On Materials and Courses

- **TCM Security Free Learning Series**

TCM Security provides a free course called "Practical Ethical Hacking," complete with hands-on labs, tool demonstrations, web application hacking content, OSINT, and more. The resources closely reflect industry expectations and are openly accessible—making them a strong supplement or primary material for practical course modules. ^[3]

- **Curated OER and Tool Guides**

University sites (such as WCCnet and ASCCC OERI) and GitHub community lists offer portals filled with curated OER textbooks, labs, and reading material relevant for teaching ethical hacking, penetration testing, and computer/network security fundamentals. ^[5] ^[6] ^[2]

Free Pentesting Resources and Tools

- Platforms like Pentest-Tools, OWASP, and others aggregate lists of free and open source network and web security testing tools. These are suitable for introductory student labs and align with the hands-on focus in the description. [\[8\]](#) [\[7\]](#)

Lab Practice Platforms

- Online hands-on labs like Hack The Box, TryHackMe, and Web Security Academy (Portswigger) are widely used in OER-based courses and provide no-cost avenues for experiential student practice. [\[11\]](#)

These resources collectively cover ethical/legal foundations, hacking methods, penetration testing steps, network/web/database tool usage, and facilitate real, hands-on security experience—all at no cost to students or faculty. Yes, there are quality OER (Open Educational Resources) for college-level courses covering ethical hacking, network security testing, penetration testing, and related legal/ethical issues.

Key OER Textbooks

- **Ethical Hacking: Network Security and Penetration Testing** (Affordable Learning Georgia/USG Galileo OER)
 - This peer-reviewed, freely available textbook addresses ethical/legal questions, hacking methodology, penetration testing using industry tools, and hands-on assignments. It covers reconnaissance, network mapping, scanning, vulnerability discovery, exploitation, and reporting—using tools like Nmap, Metasploit, Wireshark, and more. [\[1\]](#) [\[2\]](#)

Supplementary Free Courses and Hands-On Materials

- **Practical Ethical Hacking (TCM Security)**
 - Free, comprehensive video series covering practical hacking, network scanning, web application testing, tool demonstration (such as Burp Suite, Nikto), and ethical/legal concepts for beginners. Materials are continuously updated for current practices and software. [\[3\]](#)
- **Pentesting and Security Tools**
 - Free online access to penetration testing tools, scanners, and vulnerability assessment suites for hands-on use in labs and assignments, including walk-throughs, guides, and virtual machine-based exercises. [\[9\]](#) [\[10\]](#) [\[7\]](#) [\[8\]](#)

Additional Curated OER and Practice Labs

- University library guides and curated lists (ASCCC OERI, WCCnet, Kennesaw State, GitHub) collect numerous free resources—textbooks, lecture slides, open courses, and lab exercises—all focused on computer/network security testing and ethical hacking skill development. [\[4\]](#) [\[6\]](#) [\[5\]](#)

These resources collectively cover every aspect of your course description and are widely used across community college and university programs for affordable, practical cybersecurity education. Yes, there are strong OER (Open Educational Resource) textbooks and materials available for a college course on introductory ethical and legal issues in network and computer security testing, with an emphasis on hacking methodology and hands-on penetration testing.

Primary OER Textbook

- **Ethical Hacking: Network Security and Penetration Testing** (Galileo/USG OER Collection)
 - This peer-reviewed, free textbook covers foundational ethical and legal principles, the hacking methodology, hands-on penetration testing using industry tools (like Nmap, Wireshark, Metasploit), and real-world labs. ^[2] ^[1]
 - Topics include reconnaissance, scanning, vulnerability testing, exploitation, post-exploitation, and reporting, as well as detailed discussion of the legal and ethical issues in cyber testing.
 - Includes lab assignments and technical exercises suitable for entry-level cybersecurity education.

Open Hands-On Labs & Courses

- **Practical Ethical Hacking (TCM Security)**
 - A free, hands-on, beginner-friendly course series that walks through practical, ethical hacking skills, tool use, OSINT, buffer overflow basics, and web application hacking. Videos and exercises are openly available and highly relevant for lab-based teaching. ^[3]

Supplementary OER Compilations

- **University OER Collections and GitHub Resource Lists**
 - Many academic libraries, faculty OER projects, and curated GitHub lists provide centralized access to OER textbooks, slides, practice assignments, and toolkits related to ethical hacking and penetration testing, aggregating materials for easy integration into courses. ^[6] ^[4] ^[5]

Free Tools for Student Practice

- Free versions of professional penetration testing tools (e.g., Nmap, Wireshark, OWASP ZAP, Burp Suite Community) are widely used in OER-based assignments and labs, with stepwise tutorials and guides available through many OER sources. ^[10] ^[7] ^[8] ^[9]

All these materials are current, comprehensive, hands-on, and freely licensed for higher education use, making them suitable for building an engaging introductory course aligned with industry standards and ethical frameworks.



2. https://libguides.wccnet.edu/oer-subjects/computer_systems_security
3. <https://academy.tcm-sec.com/p/learn-penetration-testing-free>
4. <https://campus.kennesaw.edu/faculty-staff/open-educational-resources/index.php>
5. <https://github.com/CSIRT-MU/edu-resources>
6. <https://asccc-oeri.org/open-educational-resources-and-information-technology-and-information-systems-itis/>
7. <https://pentest-tools.com/for/free>
8. https://owasp.org/www-community/Free_for_Open_Source_Application_Security_Tools
9. <https://www.redlegg.com/blog/9-fundamental-cybersecurity-tools-for-testing-network-security>
10. <https://www.bluevoyant.com/knowledge-center/penetration-testing-tools-6-free-tools-you-should-know>
11. https://www.reddit.com/r/cybersecurity/comments/127zo25/good_resources_for_learning_pentesting_manually/
12. <https://www.oreilly.com/library/view/ethical-hacking/9781098129095/>
13. <https://www.edx.org/learn/penetration-testing>
14. <https://www.cengage.com/c/hands-on-ethical-hacking-and-network-defense-4e-wilson-simpson-antill/9780357509753/>
15. <https://www.lumifycyber.com/blog/free-open-source-software-cybersecurity/>
16. <https://www.coursera.org/courses?query=ethical+hacking>
17. <https://brightsec.com/blog/security-testing-tools/>
18. <https://www.jblearning.com/catalog/productdetails/9781284248999>
19. <https://nostarch.com/ethical-hacking>
20. https://www.youtube.com/watch?v=d_kLehoodQ